

Figure 10.1 Schematic picture of a nuclear reaction.

All the *conservation laws* derived in §5.2 are applicable to nuclear reactions. These are:

(a) the conservation of total energy:	$\Delta E = 0$	(5.4)
(b) the conservation of linear momentum:	$\Delta p = 0$	(5.6)
(c) the conservation of total charge:	$\Delta Z = 0$	(5.8)
(d) the conservation of mass number:	$\Delta A = 0$	(5.9)
(e) the conservation of spin:	$\Delta I = 0$	(5.11)

"Conservation of total energy" means that the total energy of the products must equal the total energy of the reactants, i.e. $E_{\text{products}} - E_{\text{reactants}} = \Delta E = 0$. For (5.6) it should be remembered that linear momentum is a vector property; thus

$$p_1 + 0 = p_3 \cos \theta_3 + p_4 \cos \theta_4 \tag{10.1}$$

$$p_3 \sin \theta_3 = p_4 \sin \theta_4 \tag{10.2}$$

where θ_3 and θ_4 are the respective emission angles, see Fig. 10.1. As pointed out earlier, some of these conservation laws (e.g. (d) and (e)) are not always obeyed in high energy reactions in which new elementary particles may be formed. The kinetic energy equation $E_{\text{kin}} = \frac{1}{2}mv^2$ (3.6) was deduced by Newton in 1687 assuming that the mass of the particle was independent of its velocity. We have seen that this is not true at high particle velocities (cf. Fig. 5.2). For example the relativistic mass increase of a bombarding proton is about 1% at a kinetic energy of 10 MeV. In all dynamic relations involving moving particles, the relativistic mass must be used. This is particularly important in Ch. 16, where we discuss the effect of the acceleration of charged particles to high energies in nuclear particle accelerators.

10.2. THE MASS ENERGY

As for radioactive decay, the energy of a nuclear reaction is given by its Q-value (4.10, 5.14):

$$Q\,(\text{MeV}) = -931.5\,\Delta\,M^0(\text{u}) \tag{10.3}$$

where

$$\Delta M^0 = M_3^0 + M_4^0 - M_1^0 - M_2^0 \tag{10.4}$$

The reaction energy may thus be computed from the rest masses of the reactants and products. If mass disappears in the reaction ($\Delta M^0 < 0$), energy is released: the reaction is said to be *exoergic*, and Q is positive. For $Q < 0$ the reaction is *endoergic* and $\Delta M^0 > 0$. (For comparison, in chemistry a negative value of ΔH is associated with exothermic reactions.) Tables of Q-values, especially for light projectiles, can be found in several literature sources.

The relativistic mass equation (5.26)

$$E_{\text{kin}} = \left(m - m^0\right)\mathbf{c}^2 \tag{10.5}$$

can be separated into two terms if we define

$$E_{\text{mass}}^0 \equiv m^0\mathbf{c}^2 \tag{10.6}$$

and

$$E_{\text{tot}} \equiv m\mathbf{c}^2 \tag{10.7}$$

Then

$$E_{\text{tot}} = E_{\text{kin}} + E_{\text{mass}}^0 \tag{10.8}$$

We will call E_{mass}^0 the *mass energy*, which is independent of the kinetic energy of the particle. E_{mass}^0 is *potential* energy, and can in principle be converted into any other energy form because (10.6) is a form of Einstein's mass-energy relation. E_{mass}^0 is closely related to the nuclear binding energy (cf. §4.4).

In the previous section we pointed out that for $Q > 0$ the reaction is exoergic and, as a consequence of that, mass has to disappear ($\Delta M^0 < 0$). If the total energy of the system is constant (conservation rule (a)), it becomes obvious from (10.8) that when E_{mass}^0 decreases E_{kin} must increase. Thus the products of the exoergic nuclear reaction have a higher kinetic energy than the reactants.

10.3. THE COULOMB BARRIER

Equation (10.8) is a special case of a more general equation applicable to all systems of particles:

$$E_{\text{tot}} = E_{\text{kin}} + E_{\text{pot}} \tag{10.9}$$

where E_{kin} = translational, rotational, vibrational, etc., energy and E_{pot} = mass energy, gravitational, electrostatic energy, surface energy, chemical binding energy, etc. This is the total energy referred to in (5.2).